ESAT PRACTICE TEST SERIES

ESAT 2024-25 Recall Mock

Mathematics 1 · Mathematics 2 · Physics

- 27 multiple-choice questions per module
- 40 minutes per module
- Non-calculator
- Hard, recall-informed original practice with full worked solutions

SOLUTION BOOK

ThrivingScholars

ANSWER KEY

ESAT 2024-25 Recall Mock - Mathematics 1

27 hard, recall-informed Mathematics 1 questions with complete worked solutions.

Construction verification: the summary key, marked option, card answer and worked-solution answer have been matched for every question. This is original, unofficial ESAT-style practice informed by the published specification and broad public candidate impressions; it is not an official UAT-UK paper or a reproduction of live-test material.

Mathematics 1 answer key

Q1 **C**	Q2 **A**	Q3 **D**	Q4 **E**	Q5 **D**	Q6 **B**	Q7 **B**	Q8 **C**	Q9 **A**
Q10 **E**	Q11 **C**	Q12 **B**	Q13 **E**	Q14 **E**	Q15 **A**	Q16 **D**	Q17 **A**	Q18 **B**
Q19 **D**	Q20 **A**	Q21 **D**	Q22 **C**	Q23 **E**	Q24 **B**	Q25 **D**	Q26 **E**	Q27 **C**

Mathematics 1

27 hard, recall-informed Mathematics 1 questions with complete worked solutions.

M1-01

Ratio and replacement

Answer **C**

Question 1

A tank contains an alloy in which copper and zinc are in the mass ratio $5 : 3$. Exactly $8\ \mathrm{kg}$ of the well-mixed alloy is removed and replaced by $8\ \mathrm{kg}$ of pure zinc. The new copper-to-zinc ratio is $3 : 2$.

What was the original mass of alloy in the tank?

- **A** $80\ \mathrm{kg}$
- **B** $192\ \mathrm{kg}$
- **C** $200\ \mathrm{kg}$
- **D** $\frac{200}{9}\ \mathrm{kg}$
- **E** $320\ \mathrm{kg}$

WORKED SOLUTION

Let the original mass be M kg. Removing 8 kg of the well-mixed alloy leaves

$$\text{copper} = \frac{5}{8}(M-8), \qquad \text{zinc} = \frac{3}{8}(M-8).$$

After 8 kg of zinc is added,

$$\frac{\frac{5}{8}(M-8)}{\frac{3}{8}(M-8)+8} = \frac{3}{2}.$$

Hence $10(M-8) = 9(M-8) + 192$, so $M = 200$.

Therefore the correct option is C.

Trap check: Option B ignores the removed alloy before the zinc is added; option A treats the removal as changing the copper-to-zinc ratio even though a well-mixed sample preserves it; option D reverses the final ratio; option E treats the removed mass as pure copper.

M1-02

HCF and LCM

Answer **A**

Question 2

For positive integers a and b, the highest common factor of a and b is 6, and their lowest common multiple is 180.

How many *ordered* pairs (a, b) satisfy these conditions?

- **A** 8
- **B** 4
- **C** 6
- **D** 3
- **E** 16

WORKED SOLUTION

Write $a = 6x$ and $b = 6y$, where x and y are coprime. Then

$$6xy = 180 \quad \Longrightarrow \quad xy = 30 = 2 \cdot 3 \cdot 5.$$

Each prime factor must be assigned wholly to either x or y. There are $2^3 = 8$ assignments, and order matters.

Therefore the correct option is A.

Trap check: Option B counts unordered pairs, C omits the assignments with one reduced factor equal to 1, D merely counts the distinct primes, and E incorrectly includes negative ordered pairs despite the positivity condition.

M1-03

Rates from a changing journey

Answer **D**

Question 3

A vehicle travels a route of length D. On one day it covers the first third of the route at speed v and the remainder at speed $2v$, taking 48 minutes in total.

On another day it covers the first half at speed $4v/3$, stops for 6 minutes, and then covers the second half at constant speed kv. The total time is again 48 minutes.

What is k?

A 2

B $\frac{9}{4}$

C $\frac{5}{2}$

D $\frac{12}{5}$

E $\frac{8}{3}$

WORKED SOLUTION

The first journey gives

$$\frac{D/3}{v} + \frac{2D/3}{2v} = \frac{2D}{3v} = 48,$$

so $D/v = 72$ minutes. On the second day the first half takes

$$\frac{D/2}{4v/3} = \frac{3}{8}\frac{D}{v} = 27 \text{ minutes.}$$

After the six-minute stop, 15 minutes remain. Therefore

$$\frac{D/2}{kv} = \frac{36}{k} = 15, \qquad k = \frac{12}{5}.$$

Therefore the correct option is D.

Trap check: Do not average segment speeds without checking the associated times; here the arithmetic mean happens to agree because the two segment times are equal.

M1-04

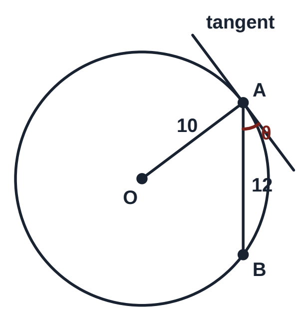
Circle geometry and trigonometry

Answer **E**

Question 4

A circle has centre O, radius 10, and chord AB of length 12. A tangent is drawn at A. Let θ be the acute angle between the tangent and chord AB.

What is $\tan\theta$?



A $\frac{4}{3}$

B $\frac{3}{5}$

C $\frac{4}{5}$

D $\frac{5}{4}$

E $\frac{3}{4}$

WORKED SOLUTION

Let M be the midpoint of AB. The perpendicular from the centre to a chord bisects it, so $AM = 6$. Hence

$$OM = \sqrt{10^2 - 6^2} = 8.$$

If $\alpha = \angle OAB$, then $\tan\alpha = 8/6 = 4/3$. Since the tangent is perpendicular to OA, $\theta = 90^\circ - \alpha$, and therefore

$$\tan\theta = \frac{1}{\tan\alpha} = \frac{3}{4}.$$

Therefore the correct option is E.

Trap check: The value 4/3 is $\tan\angle OAB$, not the tangent–chord angle requested.

M1-05

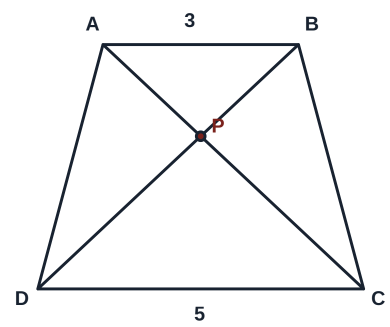
Similarity and area

Answer **D**

Question 5

In trapezium $ABCD$, $AB \parallel CD$. The diagonals AC and BD meet at P, and $AB : CD = 3 : 5$. The area of triangle APB is 27.

What is the area of triangle CPD?



A 45

B 125

C $\frac{243}{25}$

D 75

E 192

WORKED SOLUTION

Triangles APB and CPD are similar: they have equal vertically opposite angles at P, and corresponding angles from $AB \parallel CD$. Their linear scale factor is $3 : 5$, so their areas are in the ratio $9 : 25$.

$$27 : \text{area}(CPD) = 9 : 25.$$

Therefore $\text{area}(CPD) = 27 \times 25/9 = 75$.

Therefore the correct option is D.

Trap check: Option A uses the linear scale factor, B cubes it, C reverses the area factor, and E gives the area of the whole trapezium rather than triangle CPD.

M1-06

Quadratic sequences

Answer **B**

Question 6

The sequence begins

$$7,\ 14,\ 25,\ 40,\ldots$$

and has constant second differences. Let n be the smallest index for which the nth term exceeds 500, and let e be the amount by which that term exceeds 500.

Which pair (n, e) is correct?

- **A** $(15, 31)$
- **B** $(16, 32)$
- **C** $(16, 31)$
- **D** $(15, 32)$
- **E** $(17, 69)$

WORKED SOLUTION

The first differences are $7, 11, 15, \ldots$, so the second difference is 4. Fitting $u_n = 2n^2 + bn + c$ gives

$$u_n = 2n^2 + n + 4.$$

Now $u_{15} = 469$, whereas $u_{16} = 532$. Thus $n = 16$, and the excess is $e = 532 - 500 = 32$.

Therefore the correct option is B.

Trap check: A and D stop one term too early; C subtracts incorrectly after finding the correct index; E moves to the next term after the threshold has already been crossed.

M1-07

Bounds in a quotient

Answer **B**

Question 7

A sample has recorded mass $2.400\ \mathrm{kg}$, correct to the nearest $0.010\ \mathrm{kg}$, and recorded volume $800\ \mathrm{cm}^3$, correct to the nearest $10\ \mathrm{cm}^3$.

If its density is $\rho\ \mathrm{g\ cm}^{-3}$, which statement gives the exact bounds for ρ?

A $\frac{479}{159} \le \rho < \frac{481}{161}$

B $\frac{479}{161} < \rho < \frac{481}{159}$

C $\frac{480}{161} \le \rho < \frac{480}{159}$

D $\frac{2395}{800} \le \rho < \frac{2405}{800}$

E $2.995 \le \rho < 3.005$

WORKED SOLUTION

In grams and cubic centimetres,

$$2395 \le m < 2405, \qquad 795 \le V < 805.$$

For positive quantities, m/V is smallest as m takes its lower bound and V approaches its excluded upper bound; it is greatest as m approaches its excluded upper bound and V takes its lower bound. Hence

$$\frac{2395}{805} < \rho < \frac{2405}{795}.$$

Dividing numerator and denominator by 5 gives

$$\frac{479}{161} < \rho < \frac{481}{159}.$$

Therefore the correct option is B.

Trap check: The lower endpoint is also excluded: the recorded-volume interval has $V < 805$, so density can approach $2395/805$ but cannot equal it.

M1-08

Conditional probability

Answer **C**

Question 8

A bag contains 4 red, 3 blue and 2 green counters. Two counters are drawn at random without replacement.

Given that the counters are different colours, what is the probability that one of them is green?

A $\frac{5}{13}$

B $\frac{6}{13}$

C $\frac{7}{13}$

D $\frac{7}{18}$

E $\frac{14}{27}$

WORKED SOLUTION

Count unordered pairs. The number of different-colour pairs is

$$4 \cdot 3 + 4 \cdot 2 + 3 \cdot 2 = 26.$$

Of these, the pairs involving green number $4 \cdot 2 + 3 \cdot 2 = 14$. Hence the conditional probability is

$$\frac{14}{26} = \frac{7}{13}.$$

Therefore the correct option is C.

Trap check: The denominator must contain only different-colour pairs, not all $\binom{9}{2}$ pairs.

M1-09

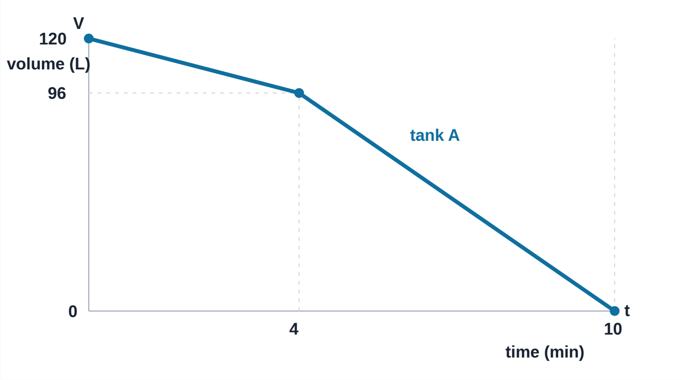
Piecewise linear graphs

Answer **A**

Question 9

The graph shows the volume V litres of water remaining in tank A, t minutes after draining begins. At $t = 0$, an empty tank B starts filling at a constant rate of 12 litres per minute.

At what time do the two tanks contain the same volume?



A $\frac{40}{7}$ min

B 5 min

C 6 min

D $\frac{20}{3}$ min

E 8 min

WORKED SOLUTION

At $t = 4$, tank A contains 96 L. From $t = 4$ to $t = 10$, its volume falls by 96 L in 6 min, so the rate is 16 L/min. On this segment,

$$V_A = 96 - 16(t - 4) = 160 - 16t.$$

Tank B contains $V_B = 12t$. Equating volumes gives

$$160 - 16t = 12t \quad \Longrightarrow \quad t = \frac{40}{7}.$$

This lies between 4 and 10, so the correct segment was used.

Therefore the correct option is A.

Trap check: The first segment's rule gives a time outside its interval; the change of gradient at 4 minutes matters.

M1-10

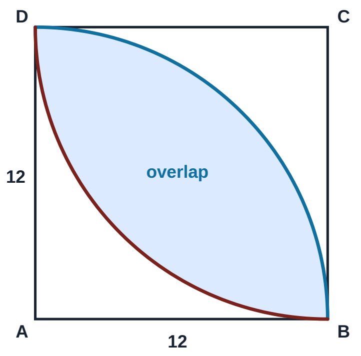
Exact area and sectors

Answer **E**

Question 10

A square $ABCD$ has side length 12. Inside it, quarter-circles of radius 12 are drawn with centres A and C. The shaded lens is the region lying inside both quarter-circles.

What is the exact area of the shaded region?



A $36\pi - 72$

B 72π

C $144 - 36\pi$

D $72\pi - 72$

E $72\pi - 144$

WORKED SOLUTION

Each arc BD subtends 90° at its centre because adjacent sides of the square are perpendicular. One half of the lens is a 90° sector of radius 12, less a right triangle with legs 12:

$$\frac{1}{4}\pi(12)^2 - \frac{1}{2}(12)(12) = 36\pi - 72.$$

The lens has two congruent halves, so its area is

$$2(36\pi - 72) = 72\pi - 144.$$

Therefore the correct option is E.

Trap check: A uses only half the lens, B subtracts no triangles, C takes an incorrect square complement, and D subtracts only one of the two right triangles.

M1-11

Conjugate surds and powers

Answer **C**

Question 11

Let $u = \sqrt{3} + \sqrt{2}$ and $v = \sqrt{3} - \sqrt{2}$.

What is the value of

$$\frac{u^6 + v^6}{u^2 + v^2}\ ?$$

- **A** 10
- **B** 94
- **C** 97
- **D** 100
- **E** 103
- **F** 970

WORKED SOLUTION

Since $uv = 1$ and $u + v = 2\sqrt{3}$,

$$u^2 + v^2 = (u+v)^2 - 2uv = 10.$$

Using $a^3 + b^3 = (a+b)^3 - 3ab(a+b)$ with $a = u^2$, $b = v^2$,

$$u^6 + v^6 = 10^3 - 3(uv)^2(10) = 1000 - 30 = 970.$$

The required quotient is $970/10 = 97$.

Therefore the correct option is C.

Trap check: Option A stops at $u^2 + v^2$, B incorrectly uses $(uv)^2 = 2$, D omits the cubic cross-term, and E changes its sign.

M1-12

Coordinate geometry

Answer **B**

Question 12

The point P lies on the line $y = 2x + 1$ and is equidistant from $A = (0, 4)$ and $B = (6, 0)$. The origin is O.

What is OP^2?

- **A** 169
- **B** 218
- **C** 196
- **D** 224
- **E** 49

WORKED SOLUTION

Write $P = (x, 2x + 1)$. Equating squared distances avoids square roots:

$$x^2 + (2x - 3)^2 = (x - 6)^2 + (2x + 1)^2.$$

The quadratic terms cancel, leaving $-12x + 9 = -8x + 37$. Hence $-4x = 28$, so $x = -7$. Therefore $P = (-7, -13)$, and

$$OP^2 = (-7)^2 + (-13)^2 = 49 + 169 = 218.$$

Therefore the correct option is B.

Trap check: A uses only the square of the y-coordinate, C omits x^2, D introduces a cross-term as if calculating $(x + y)^2$, and E reports only x^2.

M1-13

Restricted sample spaces

Answer **E**

Question 13

Two distinct cards are selected at random from cards numbered 1 to 8. It is known that their product is even and their sum is **not** divisible by 3.

What is the probability that their product is divisible by 4?

A $\frac{9}{22}$

B $\frac{1}{2}$

C $\frac{5}{14}$

D $\frac{7}{11}$

E $\frac{9}{14}$

WORKED SOLUTION

Use unordered pairs. There are 22 pairs with an even product. Of the 10 pairs whose sum is divisible by 3, two have odd product, so eight are excluded from those 22. The conditioned sample space therefore has $22 - 8 = 14$ pairs.

The nine admissible pairs with product divisible by 4 are

$$(2,6), (2,8), (1,4), (3,4), (4,6), (4,7), (6,8), (3,8), (5,8).$$

The required probability is $9/14$.

Therefore the correct option is E.

Trap check: A uses the 22 even-product pairs as the denominator and ignores the sum condition; B assumes an unsupported half-and-half split; C counts only a restricted subset centred on the cards 4 and 6; D counts products divisible by 4 before enforcing the sum condition.

M1-14

Divisors with overlapping conditions

Answer **E**

Question 14

Let

$$N = 2^4 \cdot 3^3 \cdot 5^2.$$

How many positive divisors of N are divisible by exactly one of 12 and 18?

- **A** 9
- **B** 6
- **C** 18
- **D** 33
- **E** 15
- **F** 51

WORKED SOLUTION

Divisors divisible by $12 = 2^2 \cdot 3$ number

$$(3)(3)(3) = 27.$$

Divisors divisible by $18 = 2 \cdot 3^2$ number

$$(4)(2)(3) = 24.$$

Those divisible by both are divisible by $\operatorname{lcm}(12, 18) = 36 = 2^2 3^2$, giving

$$(3)(2)(3) = 18.$$

To count divisors in exactly one set, subtract the intersection twice:

$$27 + 24 - 2(18) = 15.$$

Therefore the correct option is E.

Trap check: Options A and B count divisors in only one named set separately, C counts the intersection, and D is the ordinary inclusion–exclusion total for at least one condition.

M1-15

Multi-stage ratio

Answer **A**

Question 15

Containers A, B, C initially hold liquid in the ratio $1:3:5$. Half of the liquid in A is transferred to B. Then half of the resulting amount in B is transferred to C. Finally, C contains 100 litres more than A.

How much liquid was there originally in total?

- **A** $144\ \mathrm{L}$
- **B** $200\ \mathrm{L}$
- **C** $150\ \mathrm{L}$
- **D** $180\ \mathrm{L}$
- **E** $108\ \mathrm{L}$

WORKED SOLUTION

Let the initial amounts be $k, 3k, 5k$. After the first transfer,

$$A = \frac{1}{2}k, \qquad B = \frac{7}{2}k.$$

Half of this new amount in B is $7k/4$, so

$$B = \frac{7}{4}k, \qquad C = 5k + \frac{7}{4}k = \frac{27}{4}k.$$

Thus $C - A = (27/4 - 1/2)k = 25k/4 = 100$, giving $k = 16$. The original total was $9k = 144$ L.

Therefore the correct option is A.

Trap check: B ignores the second transfer, C transfers half of B's initial amount, D compares the final amounts in C and B, and E reports C's final amount rather than the original total.

M1-16

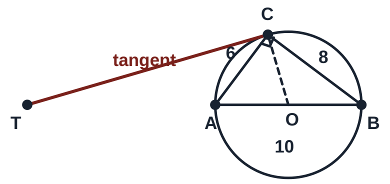
Tangent and coordinate geometry

Answer **D**

Question 16

Triangle ABC is right-angled at C, with $AC = 6$, $BC = 8$ and $AB = 10$. The circle with diameter AB is shown. The tangent to the circle at C meets the line AB, extended beyond A, at T.

What is CT?



A $\frac{60}{7}$

B 12

C 15

D $\frac{120}{7}$

E 20

WORKED SOLUTION

Place $A = (0, 0)$ and $B = (10, 0)$. The 6-8-10 triangle gives $C = (18/5, 24/5)$, and the centre is $O = (5, 0)$. The gradient of OC is

$$\frac{24/5}{18/5 - 5} = -\frac{24}{7}.$$

The tangent is perpendicular to OC, so its gradient is $7/24$. Its equation through C is

$$y - \frac{24}{5} = \frac{7}{24}\left(x - \frac{18}{5}\right).$$

Setting $y = 0$ gives $T = (-90/7, 0)$. Therefore

$$CT = \sqrt{\left(\frac{18}{5} + \frac{90}{7}\right)^2 + \left(\frac{24}{5}\right)^2} = \frac{120}{7}.$$

Therefore the correct option is D.

Trap check: The tangent is perpendicular to the radius OC, not to either side of triangle ABC.

M1-17

Similarity and an inscribed square

Answer **A**

Question 17

A right-angled triangle has perpendicular sides 9 and 12. A square is placed with one side on the hypotenuse and the opposite side parallel to it, with the other two vertices lying on the perpendicular sides.

What is the side length of the square?

- **A** $\frac{180}{37}$
- **B** $\frac{36}{7}$
- **C** $\frac{60}{11}$
- **D** $\frac{72}{13}$
- **E** 6

WORKED SOLUTION

The hypotenuse has length 15, and the altitude from the right angle to the hypotenuse is

$$h = \frac{2 \times (\frac{1}{2})(9)(12)}{15} = \frac{36}{5}.$$

If the square side is s, the small triangle above it is similar to the original triangle. Its scale factor is $(h - s)/h$, so its base, which is a side of the square, satisfies

$$s = 15\frac{h - s}{h}.$$

Thus $s(h + 15) = 15h$, and with $h = 36/5$,

$$s = \frac{15(36/5)}{15 + 36/5} = \frac{180}{37}.$$

Therefore the correct option is A.

Trap check: This is not the standard square placed at the right-angle vertex; here a side of the square lies on the hypotenuse.

M1-18

Statistics and probability

Answer **B**

Question 18

A data set contains only the values shown.

Value	1	2	3	4	5
Frequency	2	a	4	b	2

There are 20 observations and their mean is 3.1. Two observations are selected at random without replacement.

What is the probability that they have equal values?

A $\frac{39}{200}$

B $\frac{39}{190}$

C $\frac{31}{190}$

D $\frac{39}{380}$

E $\frac{3}{10}$

WORKED SOLUTION

The total frequency gives $a + b = 12$. The total of all observations is $20(3.1) = 62$, so

$$2 + 2a + 12 + 4b + 10 = 62 \quad \Longrightarrow \quad a + 2b = 19.$$

Hence $b = 7$ and $a = 5$. The number of unordered equal-value pairs is

$$\binom{2}{2} + \binom{5}{2} + \binom{4}{2} + \binom{7}{2} + \binom{2}{2} = 1 + 10 + 6 + 21 + 1 = 39.$$

There are $\binom{20}{2} = 190$ unordered pairs in total, so the probability is $39/190$.

Therefore the correct option is B.

Trap check: Using 20^2 as the denominator models selection with replacement and also permits selecting the same observation twice.

M1-19

Successive variable percentages

Answer **D**

Question 19

A price is increased by $p\%$, where $p > 0$. The increased price is then reduced by $(p+10)\%$. The final price is 12% below the original price.

What is p?

- **A** -20
- **B** 12
- **C** 22
- **D** 10
- **E** 20

WORKED SOLUTION

Let $x = p/100$. The two multipliers give

$$(1+x)(0.9-x) = 0.88.$$

Expanding and simplifying,

$$0.9 - 0.1x - x^2 = 0.88 \implies x^2 + 0.1x - 0.02 = 0.$$

Hence $(x-0.1)(x+0.2) = 0$. Since $p > 0$, $x = 0.1$, so $p = 10$.

Therefore the correct option is D.

Trap check: A is the rejected algebraic root, B copies the final percentage, C adds 12 and 10, and E neglects the product term in the two successive multipliers.

M1-20

Prime exponents

Answer **A**

Question 20

What is the smallest positive integer n for which both $72n$ is a perfect square and $50n$ is a perfect cube?

- **A** 20000
- **B** 5000
- **C** 10000
- **D** 16000
- **E** 40000

WORKED SOLUTION

Write $n = 2^a 3^b 5^c$ with the least possible non-negative exponents. Since $72 = 2^3 3^2$, the exponents in $72n$ must be even. Since $50 = 2 \cdot 5^2$, the exponents in $50n$ must be multiples of 3. The least simultaneous choices are

$$a = 5, \qquad b = 0, \qquad c = 4.$$

Thus $n = 2^5 \cdot 5^4 = 32 \cdot 625 = 20000$.

Therefore the correct option is A.

Trap check: The same n must satisfy both exponent conditions simultaneously; satisfying only the square condition is insufficient.

M1-21

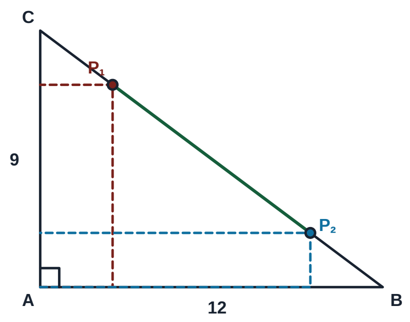
Coordinate geometry and quadratics

Answer **D**

Question 21

Triangle ABC is right-angled at A, with $AB = 12$ and $AC = 9$. For a point P on the hypotenuse, perpendiculars from P to AB and AC form a rectangle having A and P as opposite vertices.

There are two positions, P_1 and P_2, for which the rectangle has area 18. What is P_1P_2?



A $3\sqrt{3}$

B $4\sqrt{3}$

C 5

D $5\sqrt{3}$

E $7\sqrt{3}$

WORKED SOLUTION

Place $A = (0,0)$, $B = (12,0)$ and $C = (0,9)$. If $P = (x,y)$, the hypotenuse equation is $y = 9 - 3x/4$. The rectangle area condition gives

$$x\left(9 - \frac{3}{4}x\right) = 18 \quad \Longrightarrow \quad x^2 - 12x + 24 = 0.$$

The two roots are $6 \pm 2\sqrt{3}$, so the horizontal separation of P_1, P_2 is $4\sqrt{3}$. Their vertical separation is $(3/4)(4\sqrt{3}) = 3\sqrt{3}$. Hence

$$P_1P_2 = \sqrt{(4\sqrt{3})^2 + (3\sqrt{3})^2} = 5\sqrt{3}.$$

Therefore the correct option is D.

Trap check: The horizontal root difference is only 4√3; P_1P_2 also includes the vertical separation.

M1-22

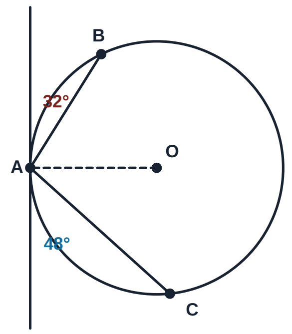
Circle theorems and tangents

Answer **C**

Question 22

At a point A on a circle, a tangent is drawn. Chords AB and AC lie on opposite sides of the radius through A. The acute angles between the tangent and the chords AB and AC are 32° and 48°, respectively.

What is the smaller angle BOC, where O is the centre?



A 100°

B 80°

C 160°

D 200°

E 40°

WORKED SOLUTION

By the tangent–chord theorem,

$$\angle ACB = 32^\circ, \qquad \angle ABC = 48^\circ.$$

Therefore $\angle BAC = 180^\circ - 32^\circ - 48^\circ = 100^\circ$. The arc BC not containing A has central angle 200°. The smaller angle at the centre is therefore

$$360^\circ - 200^\circ = 160^\circ.$$

Therefore the correct option is C.

Trap check: A gives the inscribed angle, B adds the two tangent–chord angles without doubling, D selects the reflex central angle, and E doubles their difference.

M1-23

Sequences and algebraic structure

Answer **E**

Question 23

An increasing arithmetic sequence has positive integer terms. Its terms satisfy

$$u_6^2 = u_3u_{11} \qquad \text{and} \qquad u_{11} - u_3 = 48.$$

What is the first term of the sequence?

A 6

B 9

C 12

D 18

E 15

WORKED SOLUTION

Let the first term be a and common difference be $d > 0$. Then

$$u_3 = a + 2d, \qquad u_6 = a + 5d, \qquad u_{11} = a + 10d.$$

The difference condition gives $8d = 48$, so $d = 6$. Expanding the other condition,

$$(a + 5d)^2 = (a + 2d)(a + 10d)$$

$$a^2 + 10ad + 25d^2 = a^2 + 12ad + 20d^2,$$

so $5d^2 = 2ad$. Since $d > 0$, $a = 5d/2 = 15$.

Therefore the correct option is E.

Trap check: The identity is multiplicative, not the ordinary arithmetic-mean relation for u_3, u_6 and u_{11}.

M1-24

Conditional probability by structure

Answer **B**

Question 24

Two fair six-sided dice are thrown.

Given that the product of the two scores is divisible by 6, what is the probability that their sum is prime?

A $\frac{7}{15}$

B $\frac{8}{15}$

C $\frac{1}{2}$

D $\frac{3}{5}$

E $\frac{2}{9}$

WORKED SOLUTION

The 15 ordered outcomes whose product is divisible by 6 are

$$(1,6);\ (2,3),(2,6);\ (3,2),(3,4),(3,6);\ (4,3),(4,6);\ (5,6);\ (6,1),(6,2),(6,3),(6,4),(6,5),(6,6).$$

Of these, the eight outcomes

$$(1,6),(6,1),(2,3),(3,2),(3,4),(4,3),(5,6),(6,5)$$

have prime sum. The required probability is therefore $8/15$.

Therefore the correct option is B.

Trap check: A misses one favourable ordered outcome, C counts unordered pairs, D includes one non-prime sum, and E uses the eight favourable outcomes over the original 36-outcome sample space.

M1-25

Congruences with an exclusion

Answer **D**

Question 25

The arithmetic sequence

$$5, 12, 19, 26, \dots$$

contains terms that are divisible by 11 but not divisible by 3. What is the position of the 15th such term?

- **A** 214
- **B** 225
- **C** 236
- **D** 247
- **E** 258
- **F** 159

WORKED SOLUTION

The nth term is $7n-2$. Divisibility by 11 requires

$$7n \equiv 2 \pmod{11},$$

so $n \equiv 5 \pmod{11}$. Write $n = 5 + 11k$, where $k \geq 0$. The term is divisible by 3 precisely when $n \equiv 2 \pmod{3}$. Since

$$5 + 11k \equiv 2 + 2k \pmod{3},$$

this happens when $2k \equiv 0 \pmod{3}$, so $k \equiv 0 \pmod{3}$. These values must be excluded. The allowed values of k are therefore

$$1, 2, 4, 5, 7, 8, \dots$$

There are two allowed values in each block of three. The 15th allowed value is $k = 22$, giving

$$n = 5 + 11(22) = 247.$$

Indeed, $7(247) - 2 = 1727 = 11 \times 157$, which is not divisible by 3.

Therefore the correct option is D.

Trap check: C excludes the wrong residue class of k; F counts the 15th multiple-of-11 candidate without any exclusion; A, B and E are one-block or one-candidate counting shifts.

M1-26

Recurrence and remainders

Answer **E**

Question 26

A positive real number x satisfies

$$x + \frac{1}{x} = 3.$$

For positive integers n, define

$$A_n = x^n + \frac{1}{x^n}.$$

What is the remainder when A_{2025} is divided by 10?

A 2

B 3

C 5

D 7

E 8

WORKED SOLUTION

Multiplying A_n by $x + x^{-1} = 3$ gives

$$3A_n = A_{n+1} + A_{n-1},$$

so $A_{n+1} = 3A_n - A_{n-1}$. Also $A_0 = 2$ and $A_1 = 3$. Working modulo 10,

$$A_0, A_1, A_2, A_3, A_4, A_5, A_6 \equiv 2, 3, 7, 8, 7, 3, 2.$$

The pair $(A_6, A_7) \equiv (2, 3)$ repeats the initial pair, so the remainders have period 6. Since 2025 leaves remainder 3 on division by 6,

$$A_{2025} \equiv A_3 \equiv 8 \pmod{10}.$$

Therefore the correct option is E.

Trap check: Track a pair of consecutive remainders to justify the cycle; matching one term alone is not enough.

M1-27

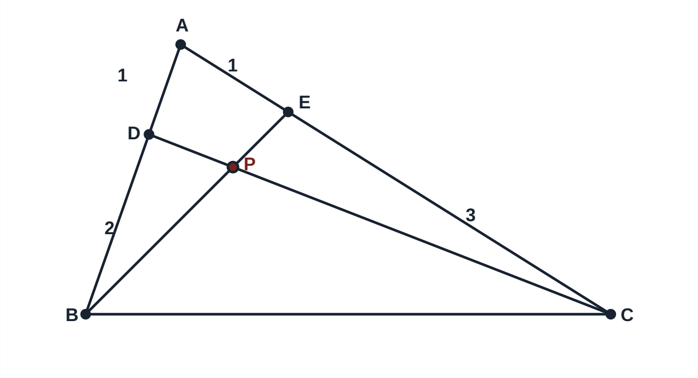
Cevians and coordinate reasoning

Answer **C**

Question 27

In triangle ABC, point D lies on AB with $AD : DB = 1 : 2$, and point E lies on AC with $AE : EC = 1 : 3$. Lines DC and EB intersect at P.

What is $EP : PB$?



- **A** $3 : 7$
- **B** $8 : 3$
- **C** $3 : 8$
- **D** $4 : 9$
- **E** $2 : 7$
- **F** $3 : 11$
- **G** $11 : 3$
- **H** $8 : 11$

WORKED SOLUTION

Choose convenient coordinates $A = (0, 0)$, $B = (3, 0)$, $C = (0, 4)$. Then $D = (1, 0)$ and $E = (0, 1)$. Parameterise P on both cevians:

$$P = D + t(C - D) = (1 - t, 4t),$$

$$P = E + s(B - E) = (3s, 1 - s).$$

Solving $1 - t = 3s$ and $4t = 1 - s$ gives $s = 3/11$. Since s is the fraction of the way from E to B,

$$EP : PB = s : (1 - s) = \frac{3}{11} : \frac{8}{11} = 3 : 8.$$

Therefore the correct option is C.

Trap check: B reverses the required ratio; F reports $EP : EB = 3 : 11$; G reverses that whole-segment ratio; H reports $PB : EB = 8 : 11$. A, D and E arise from using one of the side ratios directly or mismatching coefficients in the two line parameters.

ANSWER KEY

ESAT 2024-25 Recall Mock - Mathematics 2

27 hard, recall-informed Mathematics 2 questions with complete worked solutions.

Construction verification: the summary key, marked option, card answer and worked-solution answer have been matched for every question. This is original, unofficial ESAT-style practice informed by the published specification and broad public candidate impressions; it is not an official UAT-UK paper or a reproduction of live-test material.

Mathematics 2 answer key

Q1 **B**	Q2 **D**	Q3 **A**	Q4 **C**	Q5 **A**	Q6 **E**	Q7 **D**	Q8 **A**	Q9 **B**
Q10 **B**	Q11 **E**	Q12 **D**	Q13 **C**	Q14 **A**	Q15 **C**	Q16 **C**	Q17 **A**	Q18 **D**
Q19 **C**	Q20 **C**	Q21 **D**	Q22 **E**	Q23 **B**	Q24 **E**	Q25 **B**	Q26 **E**	Q27 **B**

Mathematics 2

27 hard, recall-informed Mathematics 2 questions with complete worked solutions.

M2-01

Geometric progressions

Answer **B**

Question 1

A geometric progression has positive terms, first term a and common ratio r. The sum of its first three terms is 14, and the sum of its first six terms is 126. What is $a + r^2$?

- **A** 4
- **B** 6
- **C** 8
- **D** 10
- **E** 14

WORKED SOLUTION

Let the sum of the first three terms be $S_3 = 14$. The next three terms are obtained by multiplying the first three by r^3, so

$$S_6 = S_3(1 + r^3).$$

Hence $126 = 14(1 + r^3)$, giving $r^3 = 8$. Positivity selects $r = 2$. Now

$$14 = a(1 + r + r^2) = 7a,$$

so $a = 2$. Therefore $a + r^2 = 2 + 4 = 6$.

M2-02

Factor and remainder theorems

Answer **D**

Question 2

The polynomial $p(x) = x^3 + ax^2 + bx + 6$ is divisible by $x - 3$. It also leaves the same remainder when divided by $x - 1$ and by $x + 2$. What is $p(2)$?

- **A** -11
- **B** -9
- **C** -5
- **D** -4
- **E** -7
- **F** -2

WORKED SOLUTION

By the factor theorem, $p(3) = 0$, so

$$27 + 9a + 3b + 6 = 0 \implies 3a + b = -11.$$

Equal remainders give $p(1) = p(-2)$:

$$7 + a + b = -2 + 4a - 2b \implies a - b = 3.$$

Solving gives $a = -2$ and $b = -5$. Therefore

$$p(2) = 8 + 4(-2) + 2(-5) + 6 = -4.$$

M2-03

Numerical methods

Answer **A**

Question 3

The table shows three values of a twice-differentiable function f.

x	0	1	3
$f(x)$	4	7	5

It is known that $f''(x) < 0$ for $0 < x < 3$. The trapezium rule is applied separately on $[0, 1]$ and $[1, 3]$. Which statement is correct?

- **A** The estimate is $\frac{35}{2}$ and is an underestimate.
- **B** The estimate is $\frac{35}{2}$ and is an overestimate.
- **C** The estimate is $\frac{23}{2}$ and is an underestimate.
- **D** The estimate is $\frac{27}{2}$ and is an underestimate.
- **E** The estimate is 23 and is an overestimate.

WORKED SOLUTION

The unequal strip widths must be retained:

$$T = \frac{1}{2}(4 + 7) + \frac{2}{2}(7 + 5) = \frac{11}{2} + 12 = \frac{35}{2}.$$

Since $f''(x) < 0$, the graph is concave down. Each chord lies below the graph, so the trapezium-rule estimate is an underestimate.

M2-04

Arithmetic progressions

Answer **C**

Question 4

For an arithmetic progression, the sum of the first 10 terms is 5, and the sum of the first 20 terms is 110. What is the sum of the positive terms among its first 30 terms?

- **A** 285
- **B** 300
- **C** 325
- **D** 315
- **E** 330

WORKED SOLUTION

For first term a and common difference d,

$$S_{10} = 5(2a + 9d) = 5, \qquad S_{20} = 10(2a + 19d) = 110.$$

Thus $2a + 9d = 1$ and $2a + 19d = 11$, so $d = 1$ and $a = -4$. The first 30 terms are therefore

$$-4, -3, -2, -1, 0, 1, 2, \ldots, 25.$$

The positive terms are 1 to 25, with sum

$$\frac{25 \cdot 26}{2} = 325.$$

M2-05

Differentiation

Answer **A**

Question 5

The curve $y = x^3 - 6x^2 + 9x + 4$ has two stationary points. The straight line joining them meets the x-axis at $(q, 0)$. What is q?

- **A** 5
- **B** 4
- **C** 6
- **D** 3
- **E** 2

WORKED SOLUTION

Differentiating gives $y' = 3(x - 1)(x - 3)$, so the stationary points are $(1, 8)$ and $(3, 4)$. The line joining them has gradient

$$\frac{4 - 8}{3 - 1} = -2,$$

so its equation is $y - 8 = -2(x - 1)$, or $y = 10 - 2x$. Setting $y = 0$ gives $q = 5$.

M2-06

Logarithms and inequalities

Answer **E**

Question 6

Solve $\log_2(x^2 - 5x + 6) > 0$.

- **A** $2 < x < 3$
- **B** $x < 2$ or $x > 3$
- **C** $\frac{5-\sqrt{5}}{2} < x < \frac{5+\sqrt{5}}{2}$
- **D** $x < \frac{5-\sqrt{5}}{2}$ or $x > 3$
- **E** $x < \frac{5-\sqrt{5}}{2}$ or $x > \frac{5+\sqrt{5}}{2}$
- **F** $x \le \frac{5-\sqrt{5}}{2}$ or $x \ge \frac{5+\sqrt{5}}{2}$
- **G** $x < \frac{5-\sqrt{5}}{2}$
- **H** $x > \frac{5+\sqrt{5}}{2}$

WORKED SOLUTION

Because the base is greater than 1, the inequality is equivalent to

$$x^2 - 5x + 6 > 1 \iff x^2 - 5x + 5 > 0.$$

The roots are $\frac{5\pm\sqrt{5}}{2}$, and the quadratic is positive outside them. On this set the logarithm's argument is automatically greater than 1, so the domain condition is satisfied.

M2-07

Trigonometric identities

Answer **D**

Question 7

An acute angle θ satisfies $\tan\theta + \sec\theta = 3$. What is $\sin 2\theta$?

- **A** $\frac{4}{5}$
- **B** $\frac{3}{5}$
- **C** $\frac{16}{25}$
- **D** $\frac{24}{25}$
- **E** $\frac{25}{24}$

WORKED SOLUTION

Use $\sec^2\theta - \tan^2\theta = 1$:

$$(\sec\theta + \tan\theta)(\sec\theta - \tan\theta) = 1.$$

Therefore $\sec\theta - \tan\theta = 1/3$. Solving the two linear equations gives $\sec\theta = 5/3$ and $\tan\theta = 4/3$. Since θ is acute, $\cos\theta = 3/5$ and $\sin\theta = 4/5$. Hence

$$\sin 2\theta = 2\sin\theta\cos\theta = \frac{24}{25}.$$

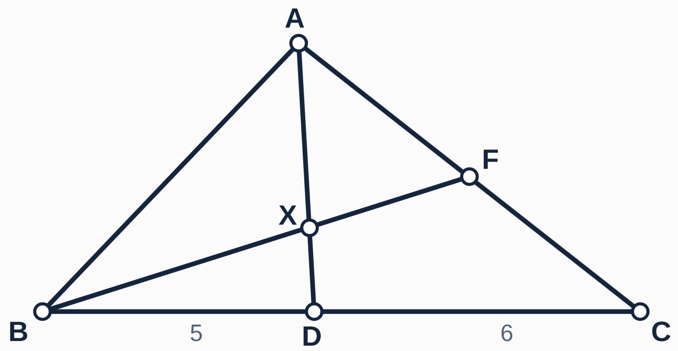
M2-08

Coordinate and ratio geometry

Answer **A**

Question 8

In triangle ABC, point D lies on BC with $BD : DC = 5 : 6$, and F is the midpoint of AC. The lines AD and BF meet at X. Find $AX : XD$.



A $11 : 5$

B $5 : 11$

C $11 : 6$

D $6 : 11$

E $16 : 5$

WORKED SOLUTION

Take A as the origin and write the position vectors of B, C as $\mathbf{b}, \mathbf{c}$. Then

$$\mathbf{d} = \frac{6\mathbf{b} + 5\mathbf{c}}{11}, \qquad \mathbf{f} = \frac{\mathbf{c}}{2}.$$

A point on AD has vector $t\mathbf{d}$, while a point on BF has vector $(1 - s)\mathbf{b} + \frac{s}{2}\mathbf{c}$. Equating coefficients gives

$$\frac{6t}{11} = 1 - s, \qquad \frac{5t}{11} = \frac{s}{2}.$$

Hence $s = \frac{10t}{11}$ and $6t = 11 - 10t$, so $t = \frac{11}{16}$. Thus $AX : XD = t : (1 - t) = 11 : 5$.

M2-09

Coordinate geometry

Answer **B**

Question 9

A straight line with positive intercepts on the coordinate axes passes through $(2,3)$. The triangle enclosed by the line and the positive coordinate axes has area 12. What is the sum of the two intercepts?

- **A** 8
- **B** 10
- **C** 12
- **D** 11
- **E** 9
- **F** 24

WORKED SOLUTION

Let the intercepts be $a,b>0$. The area condition gives $ab=24$, and the intercept form of the line gives

$$\frac{2}{a}+\frac{3}{b}=1.$$

Substituting $b=24/a$ gives $2/a+a/8=1$, so $a^2-8a+16=0$. Hence $a=4$, $b=6$, and $a+b=10$.

M2-10

Infinite geometric series

Answer **B**

Question 10

A positive geometric progression has sum to infinity 24, and the sum of its first three terms is 21. What is the sum of all terms whose positions are positive multiples of 3?

- **A** 8
- **B** $\frac{24}{7}$
- **C** $\frac{12}{7}$
- **D** 3
- **E** 6

WORKED SOLUTION

Since $S_3=S_\infty(1-r^3)$,

$$21=24(1-r^3),$$

so $r^3=\frac{1}{8}$ and, because the terms are positive, $r=\frac{1}{2}$. Then $a=24(1-r)=12$. The terms in positions $3,6,9,\dots$ form

$$ar^2+ar^5+ar^8+\cdots,$$

a geometric series with first term $ar^2=3$ and ratio $r^3=1/8$. Its sum is

$$\frac{3}{1-1/8}=\frac{24}{7}.$$

M2-11

Exponential equations

Answer **E**

Question 11

Solve $2^x + 2^{-x} = \frac{5}{2}$.

A $x = -1$ only

B $x = 0, \pm 1$

C $x = 1$ only

D $x = \pm 2$

E $x = \pm 1$

WORKED SOLUTION

Let $u = 2^x > 0$. Then $u + u^{-1} = \frac{5}{2}$, so $2u^2 - 5u + 2 = 0$. Hence $u = 2$ or $u = \frac{1}{2}$, giving $x = 1$ or $x = -1$.

M2-12

Integration

Answer **D**

Question 12

Let

$$A = \int_0^{1/2} (\sqrt{x} - x\sqrt{x})^2 \, dx, \qquad B = \int_{1/2}^{1} (\sqrt{x} - x\sqrt{x})^2 \, dx.$$

What is $A : B$?

A $5 : 11$

B $16 : 5$

C $11 : 16$

D $11 : 5$

E $5 : 16$

WORKED SOLUTION

Expand before integrating:

$$(\sqrt{x} - x\sqrt{x})^2 = x - 2x^2 + x^3.$$

An antiderivative is

$$F(x) = \frac{x^2}{2} - \frac{2x^3}{3} + \frac{x^4}{4}.$$

Hence $A = F(1/2) - F(0) = 11/192$. Also $F(1) = 1/12 = 16/192$, so

$$B = F(1) - F(1/2) = \frac{5}{192}.$$

Therefore $A : B = 11 : 5$.

M2-13

Parameters and roots

Answer **C**

Question 13

The equation $x^2 - 4x + k = 0$ has two distinct real roots α and β. Given that $\alpha^4 + \beta^4 = 194$, find k.

- **A** 31
- **B** -1
- **C** 1
- **D** 4
- **E** 32

WORKED SOLUTION

Here $\alpha + \beta = 4$ and $\alpha\beta = k$. Thus

$$\alpha^2 + \beta^2 = 16 - 2k$$

and

$$\alpha^4 + \beta^4 = (16 - 2k)^2 - 2k^2 = 194.$$

This simplifies to $k^2 - 32k + 31 = 0$, so $k = 1$ or 31. Two distinct real roots require the discriminant $16 - 4k > 0$, hence $k < 4$. Therefore $k = 1$.

M2-14

Binomial expansions

Answer **A**

Question 14

What is the coefficient of x^3 in $(1 + 2x)^5(1 - x)^3$?

- **A** -11
- **B** -13
- **C** -9
- **D** 11
- **E** -15

WORKED SOLUTION

The x^3-coefficient is the sum of the four compatible products:

$$\binom{5}{3}2^3 - \binom{5}{2}2^2\binom{3}{1} + \binom{5}{1}2\binom{3}{2} - \binom{3}{3}.$$

This is $80 - 120 + 30 - 1 = -11$.

M2-15

Modulus and parameters

Answer **C**

Question 15

For $k > 0$, the equation $|x^2 - 4x + 3| = k$ has exactly three distinct real solutions. What is the sum of those solutions?

A 4

B 2

C 6

D $4 + \sqrt{2}$

E 8

WORKED SOLUTION

Write $x^2 - 4x + 3 = (x-2)^2 - 1$. The branch $(x-2)^2 - 1 = k$ has two roots for every $k > 0$. The other branch is

$$(x-2)^2 - 1 = -k \quad \iff \quad (x-2)^2 = 1 - k.$$

It contributes two roots if $0 < k < 1$, one root if $k = 1$, and none if $k > 1$. Exactly three roots therefore requires $k = 1$. They are $2 - \sqrt{2}, 2, 2 + \sqrt{2}$, with sum 6.

M2-16

Circle geometry

Answer **C**

Question 16

The line $y = 2x + 1$ cuts the circle $x^2 + y^2 = 25$ in a chord. What is the area of the triangle whose vertices are the origin and the two endpoints of the chord?

A $\frac{\sqrt{31}}{5}$

B $\frac{4\sqrt{31}}{5}$

C $\frac{2\sqrt{31}}{5}$

D $2\sqrt{31}$

E $\sqrt{31}$

WORKED SOLUTION

The perpendicular distance from the origin to $2x - y + 1 = 0$ is $d = 1/\sqrt{5}$. If h is half the chord length, then

$$h^2 + d^2 = 5^2 \quad \Longrightarrow \quad h^2 = 25 - \frac{1}{5} = \frac{124}{5}.$$

The full chord length is $2h = 2\sqrt{124/5} = 4\sqrt{31/5}$. Taking this chord as the base and d as the perpendicular height, the triangle area is

$$\frac{1}{2}(2h)d = \sqrt{\frac{124}{5}} \cdot \frac{1}{\sqrt{5}} = \frac{2\sqrt{31}}{5}.$$

M2-17

Area and integration

Answer **A**

Question 17

For $f(x) = x^2 - 4x + 3$, evaluate

$$\frac{\displaystyle\int_0^4 |f(x)|\,dx}{\left|\displaystyle\int_0^4 f(x)\,dx\right|}.$$

- **A** 3
- **B** 2
- **C** 1
- **D** 4
- **E** $\frac{1}{3}$

WORKED SOLUTION

The curve crosses the axis at $x = 1, 3$, and is below it between those values. With

$$F(x) = \frac{x^3}{3} - 2x^2 + 3x,$$

the three unsigned region areas are each $4/3$, so

$$\int_0^4 |f(x)|\,dx = 4.$$

The signed integral is $4/3 - 4/3 + 4/3 = 4/3$. The required ratio is therefore

$$\frac{4}{4/3} = 3.$$

M2-18

Parameters and stationary points

Answer **D**

Question 18

The curve $y = x^3 - 3x^2 + kx$ has two distinct stationary points. The gradient of the straight line joining them is -1. Find k.

A $\frac{1}{2}$

B 1

C 3

D $\frac{3}{2}$

E $\frac{9}{2}$

WORKED SOLUTION

Let the stationary x-coordinates be r, s. Since

$$3x^2 - 6x + k = 0,$$

Vieta gives $r + s = 2$ and $rs = k/3$. For $f(x) = x^3 - 3x^2 + kx$, the secant gradient between r and s is

$$\frac{f(r) - f(s)}{r - s} = r^2 + rs + s^2 - 3(r + s) + k.$$

Now $r^2 + rs + s^2 = (r + s)^2 - rs = 4 - k/3$, so the gradient is $-2 + 2k/3$. Setting this equal to -1 gives $k = 3/2$. The derivative discriminant is then positive, so the stationary points are distinct.

M2-19

Composite functions

Answer **C**

Question 19

Let $f(x) = x^2 - 1$ and $g(x) = 2x + 1$. If the real solutions of $f(g(x)) = g(f(x))$ are r and s, what is $f(r) + f(s)$?

A -2

B $\frac{1}{2}$

C 1

D 2

E 3

WORKED SOLUTION

We have

$$f(g(x)) = (2x + 1)^2 - 1 = 4x^2 + 4x$$

and $g(f(x)) = 2(x^2 - 1) + 1 = 2x^2 - 1$. Hence

$$2x^2 + 4x + 1 = 0.$$

The discriminant is positive, so both roots are real. By Vieta, $r + s = -2$ and $rs = 1/2$. Therefore

$$f(r) + f(s) = r^2 + s^2 - 2 = (r + s)^2 - 2rs - 2 = 4 - 1 - 2 = 1.$$

M2-20

Trigonometric equations

Answer **C**

Question 20

The solutions of $2\sin^2 x + \sin x - 1 = 0$ for $0 \le x < 2\pi$ have sum

A 3π

B $\frac{11\pi}{4}$

C $\frac{5\pi}{2}$

D 2π

E $\frac{9\pi}{4}$

WORKED SOLUTION

Factorising gives $(2\sin x - 1)(\sin x + 1) = 0$. Thus

$$x = \frac{\pi}{6}, \quad \frac{5\pi}{6}, \quad \frac{3\pi}{2}.$$

Their sum is $\frac{5\pi}{2}$.

M2-21

Graph transformations

Answer **D**

Question 21

Let $f(x) = |x^2 - 4x + 3|$. How many real solutions does $f(|x|) = 1$ have?

A 2

B 3

C 4

D 6

E 8

F 5

G 7

H 12

WORKED SOLUTION

Put $t = |x|$, so $t \ge 0$. We need

$$|t^2 - 4t + 3| = 1.$$

The positive branch gives $(t-2)^2 = 2$, hence $t = 2 \pm \sqrt{2}$, both positive. The negative branch gives $(t-2)^2 = 0$, hence $t = 2$. Each of these three positive values of t corresponds to two values $x = \pm t$. Therefore there are 6 real solutions.

M2-22

Parameters and coordinate geometry

Answer **E**

Question 22

The line $y = kx + 1$ intersects the parabola $y = x^2 - 4x + 5$ at two points whose x-coordinates differ by 4. What is the sum of all possible values of k?

- **A** -4
- **B** $-4\sqrt{2}$
- **C** 8
- **D** 4
- **E** -8

WORKED SOLUTION

At an intersection,

$$x^2 - (k+4)x + 4 = 0.$$

If its roots are r, s, then $(r-s)^2 = (r+s)^2 - 4rs$. Here $r + s = k + 4$ and $rs = 4$, so

$$16 = (k+4)^2 - 16.$$

Hence $(k+4)^2 = 32$, giving $k = -4 \pm 4\sqrt{2}$. Their sum is -8.

M2-23

Tangents to curves

Answer **B**

Question 23

Two distinct tangents drawn from the origin touch the parabola $y = x^2 - 4x + 5$. What is the product of the x-coordinates of the two points of contact?

- **A** 0
- **B** -5
- **C** -8
- **D** 5
- **E** -4

WORKED SOLUTION

At $x = a$, the tangent has gradient $2a - 4$. Because it passes through the origin, its gradient is also $f(a)/a$. Therefore

$$a(2a-4) = a^2 - 4a + 5 \quad \Longrightarrow \quad a^2 = 5.$$

The contact coordinates have x-values $\sqrt{5}$ and $-\sqrt{5}$, whose product is -5.

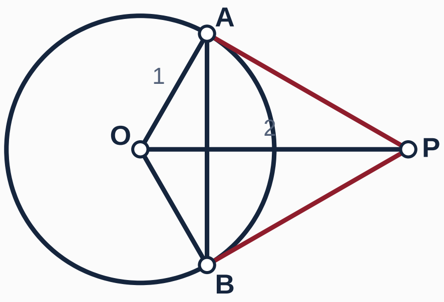
M2-24

Radian measure and circle geometry

Answer **E**

Question 24

A circle has centre O and radius 1. A point P satisfies $OP = 2$, and the tangents from P touch the circle at A and B. What is the area of the smaller segment bounded by chord AB and the minor arc AB?



A $\frac{\pi}{6} - \frac{\sqrt{3}}{8}$

B $\frac{\pi}{3} + \frac{\sqrt{3}}{4}$

C $\frac{2\pi}{3} - \frac{\sqrt{3}}{4}$

D $\frac{\pi}{3} - \frac{\sqrt{3}}{2}$

E $\frac{\pi}{3} - \frac{\sqrt{3}}{4}$

F $\frac{2\pi}{3} - \frac{\sqrt{3}}{2}$

G $\frac{\pi}{6} - \frac{\sqrt{3}}{4}$

H $\frac{\pi}{3} - \frac{\sqrt{3}}{8}$

WORKED SOLUTION

Since a radius is perpendicular to a tangent, triangle OAP is right-angled at A. Thus

$$\cos \angle AOP = \frac{OA}{OP} = \frac{1}{2},$$

so $\angle AOP = \pi/3$. By symmetry, $\angle AOB = 2\pi/3$. The sector area is $\frac{1}{2}(1)^2(2\pi/3) = \pi/3$, while

$$\text{area}(\triangle AOB) = \frac{1}{2}\sin(2\pi/3) = \frac{\sqrt{3}}{4}.$$

The required segment area is $\frac{\pi}{3} - \frac{\sqrt{3}}{4}$.

M2-25

Sine rule and area

Answer **B**

Question 25

In triangle ABC, side a is opposite angle A, and side b is opposite angle B. Given $a = 2$, $b = 2\sqrt{3}$, and $A = 30^\circ$, two non-congruent triangles are possible. What is the sum of their areas?

- **A** $\sqrt{3}$
- **B** $3\sqrt{3}$
- **C** $2\sqrt{3}$
- **D** $4\sqrt{3}$
- **E** 6

WORKED SOLUTION

By the sine rule,

$$\sin B = \frac{b \sin A}{a} = \frac{2\sqrt{3}\,(1/2)}{2} = \frac{\sqrt{3}}{2}.$$

Hence $B = 60^\circ$ or 120°, giving $C = 90^\circ$ or 30°. Since the included angle between sides a, b is C, the two areas are

$$\frac{1}{2}ab\sin 90^\circ = 2\sqrt{3}, \qquad \frac{1}{2}ab\sin 30^\circ = \sqrt{3}.$$

Their sum is $3\sqrt{3}$.

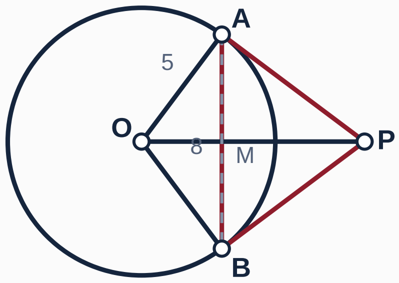
M2-26

Tangents and circle geometry

Answer **E**

Question 26

A circle has centre O and radius 5. A chord AB has length 8. The tangents at A and B meet at P. Find the area of quadrilateral $OAPB$.



A $\frac{50}{3}$

B $\frac{80}{3}$

C 25

D $\frac{200}{3}$

E $\frac{100}{3}$

WORKED SOLUTION

Let M be the midpoint of AB. Then $AM = 4$, and $OM = \sqrt{5^2 - 4^2} = 3$. Symmetry puts O, M, P on one line. Triangles OMA and OAP are similar: both are right-angled and share the angle at O. Therefore

$$\frac{OP}{OA} = \frac{OA}{OM} \quad \Longrightarrow \quad OP = \frac{25}{3}.$$

Hence $AP = \sqrt{OP^2 - OA^2} = 20/3$. Quadrilateral $OAPB$ is two congruent right triangles, so its area is

$$2\left(\frac{1}{2} \cdot 5 \cdot \frac{20}{3}\right) = \frac{100}{3}.$$

M2-27

Differential equations and area

Answer **B**

Question 27

A curve satisfies $\dfrac{dy}{dx} = 3x(x-2)$. It is tangent to the x-axis at its positive stationary point. What is the finite area enclosed by the curve and the x-axis?

A $\frac{9}{2}$

B $\frac{27}{4}$

C $\frac{11}{4}$

D 4

E 8

WORKED SOLUTION

Integrating gives $y = x^3 - 3x^2 + C$. The positive stationary point is at $x = 2$, and tangency to the x-axis means $y(2) = 0$, so $C = 4$. Thus

$$y = x^3 - 3x^2 + 4 = (x-2)^2(x+1).$$

The curve meets the axis at $x = -1$ and touches it at $x = 2$; it is non-negative between them. Hence the enclosed area is

$$\int_{-1}^{2} (x^3 - 3x^2 + 4)\, dx = \left[\frac{x^4}{4} - x^3 + 4x\right]_{-1}^{2} = 4 - \left(-\frac{11}{4}\right) = \frac{27}{4}.$$

ANSWER KEY

ESAT 2024-25 Recall Mock - Physics

27 hard, recall-informed Physics questions with complete worked solutions.

Construction verification: the summary key, marked option, card answer and worked-solution answer have been matched for every question. This is original, unofficial ESAT-style practice informed by the published specification and broad public candidate impressions; it is not an official UAT-UK paper or a reproduction of live-test material.

Physics answer key

Q1 **E**	Q2 **B**	Q3 **A**	Q4 **C**	Q5 **E**	Q6 **B**	Q7 **C**	Q8 **A**	Q9 **B**
Q10 **D**	Q11 **C**	Q12 **E**	Q13 **C**	Q14 **C**	Q15 **A**	Q16 **C**	Q17 **E**	Q18 **B**
Q19 **B**	Q20 **C**	Q21 **D**	Q22 **E**	Q23 **C**	Q24 **B**	Q25 **D**	Q26 **A**	Q27 **E**

Physics

27 hard, recall-informed Physics questions with complete worked solutions.

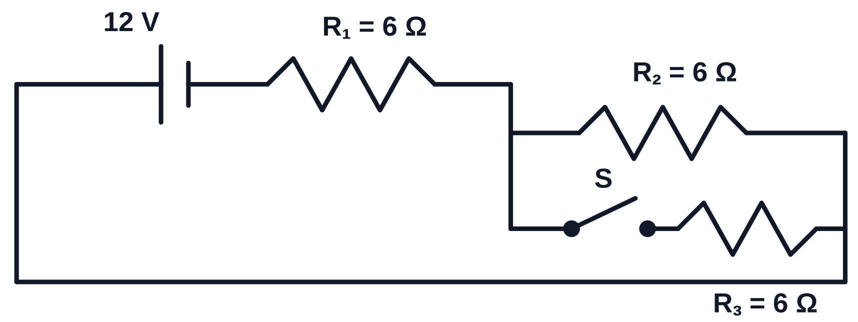
P-01

Qualitative circuit changes

Answer **E**

Question 1

Three identical $6\,\Omega$ resistors are connected to an ideal $12\ \mathrm{V}$ source as shown. Initially switch S is open; it is then closed.



Which statements about closing S are correct?

1. The current supplied by the source increases.
2. The potential difference across R_1 increases.
3. The power dissipated in R_2 decreases.

A 1 only

B 2 only

C 1 and 3 only

D 2 and 3 only

E 1, 2 and 3

WORKED SOLUTION

With S open, the total resistance is $12\,\Omega$, so the source current is $1\ \mathrm{A}$. The voltage across R_1 is $6\ \mathrm{V}$, and R_2 dissipates $6^2/6 = 6\ \mathrm{W}$.

Closing S puts R_2 and R_3 in parallel: their resistance is $3\,\Omega$, so the total is $9\,\Omega$. The source current rises to $4/3\ \mathrm{A}$, the voltage across R_1 rises to $8\ \mathrm{V}$, and the parallel branch voltage becomes $4\ \mathrm{V}$. Thus R_2's power falls to $4^2/6 = 8/3\ \mathrm{W}$. All three statements are correct, so the answer is **E.**

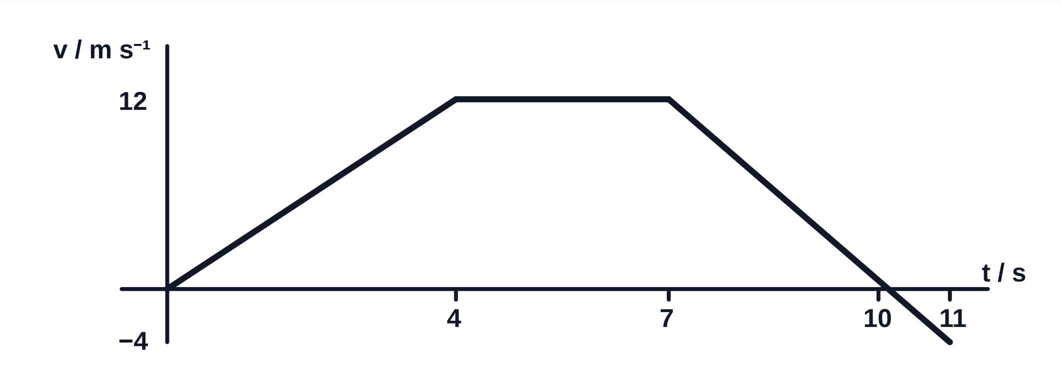
P-02

Velocity-time graphs

Answer **B**

Question 2

A particle starts from rest. It accelerates uniformly to $12\ \mathrm{m\,s^{-1}}$ in $4\ \mathrm{s}$, travels at this velocity for $3\ \mathrm{s}$, then changes velocity uniformly to $-4\ \mathrm{m\,s^{-1}}$ over the next $4\ \mathrm{s}$. What are its total displacement and total distance travelled?



A $80\ \mathrm{m},\ 80\ \mathrm{m}$

B $76\ \mathrm{m},\ 80\ \mathrm{m}$

C $80\ \mathrm{m},\ 84\ \mathrm{m}$

D $76\ \mathrm{m},\ 76\ \mathrm{m}$

E $72\ \mathrm{m},\ 76\ \mathrm{m}$

WORKED SOLUTION

Signed area gives displacement:

$$\tfrac{1}{2}(4)(12) + (3)(12) + \tfrac{1}{2}(12-4)(4) = 24 + 36 + 16 = 76\ \mathrm{m}.$$

In the final section the velocity reaches zero after $3\ \mathrm{s}$. Its distance in that section is $\frac{1}{2}(3)(12) + \frac{1}{2}(1)(4) = 20\ \mathrm{m}$. Thus total distance is $24 + 36 + 20 = 80\ \mathrm{m}$. The correct option is **B.**

P-03

Wave data and refraction

Answer **A**

Question 3

Five complete cycles of a wave pass a fixed point in medium X in $0.020\ \mathrm{s}$. Adjacent crests in X are $0.80\ \mathrm{m}$ apart. The wave then enters medium Y obliquely, where its speed is $\frac{3}{4}$ of its speed in X. Which row gives the speed, frequency and wavelength in Y, together with the direction of bending?

- **A** $150\ \mathrm{m\,s^{-1}},\ 250\ \mathrm{Hz},\ 0.60\ \mathrm{m}$; towards the normal
- **B** $150\ \mathrm{m\,s^{-1}},\ 187.5\ \mathrm{Hz},\ 0.80\ \mathrm{m}$; towards the normal
- **C** $200\ \mathrm{m\,s^{-1}},\ 250\ \mathrm{Hz},\ 0.60\ \mathrm{m}$; away from the normal
- **D** $150\ \mathrm{m\,s^{-1}},\ 250\ \mathrm{Hz},\ 0.60\ \mathrm{m}$; away from the normal
- **E** $200\ \mathrm{m\,s^{-1}},\ 250\ \mathrm{Hz},\ 0.80\ \mathrm{m}$; no change in direction

WORKED SOLUTION

The frequency is $f = 5/0.020 = 250\ \mathrm{Hz}$. In X, $v_X = f\lambda_X = 250(0.80) = 200\ \mathrm{m\,s^{-1}}$. Therefore

$$v_Y = \frac{3}{4}v_X = 150\ \mathrm{m\,s^{-1}}, \qquad \lambda_Y = \frac{v_Y}{f} = \frac{150}{250} = 0.60\ \mathrm{m}.$$

Frequency does not change at the boundary. Since the wave slows down, it bends towards the normal. The correct option is **A.**

P-04

Composite thermal conductors

Answer **C**

Question 4

Two rods P and Q of equal cross-sectional area are joined end to end. Rod P has length L and thermal conductivity k. Rod Q has length $2L$ and thermal conductivity $3k$. The free end of P is kept at $100^\circ\mathrm{C}$, and the free end of Q at $20^\circ\mathrm{C}$. Assume one-dimensional conduction: the rods' sides are thermally insulated, contact resistance at the junction is negligible, and no heat is lost to the surroundings. In steady state, the transfer rate through a rod is $kA\Delta T/L$. What is the junction temperature?

- **A** $36^\circ\mathrm{C}$
- **B** $44^\circ\mathrm{C}$
- **C** $52^\circ\mathrm{C}$
- **D** $60^\circ\mathrm{C}$
- **E** $68^\circ\mathrm{C}$
- **F** $48^\circ\mathrm{C}$

WORKED SOLUTION

The thermal resistances are proportional to $L/(kA)$. Thus

$$R_P : R_Q = 1 : \frac{2}{3} = 3 : 2.$$

The total temperature drop is $80^\circ\mathrm{C}$, so the drop across P is $\frac{3}{5}(80) = 48^\circ\mathrm{C}$. The junction is therefore at $100 - 48 = 52^\circ\mathrm{C}$. The correct option is **C.**

P-05

AC generators

Answer **E**

Question 5

A simple AC generator contains a coil of N turns rotating at angular speed ω in a uniform magnetic field and is connected to a fixed resistor. For this generator, $V_{\text{peak}} \propto N\omega$, $f \propto \omega$, and the average power in the resistor is proportional to V_{peak}^2.

The coil is replaced by one of $2N$ turns rotating at angular speed $\omega/2$, with all other factors unchanged. How do the peak voltage and the energy delivered to the resistor in one complete cycle change?

A Peak voltage doubles; energy per cycle doubles

B Peak voltage is unchanged; energy per cycle is unchanged

C Peak voltage halves; energy per cycle halves

D Peak voltage doubles; energy per cycle is four times as large

E Peak voltage is unchanged; energy per cycle doubles

WORKED SOLUTION

The peak induced voltage scales as $N\omega$, so the two changes multiply it by $2 \times \frac{1}{2} = 1$: the peak voltage is unchanged. The average power in the fixed resistor is therefore unchanged.

The frequency halves, so the period of one cycle doubles. Since energy per cycle equals average power multiplied by the period, the energy delivered in each complete cycle doubles. The correct option is **E.**

P-06

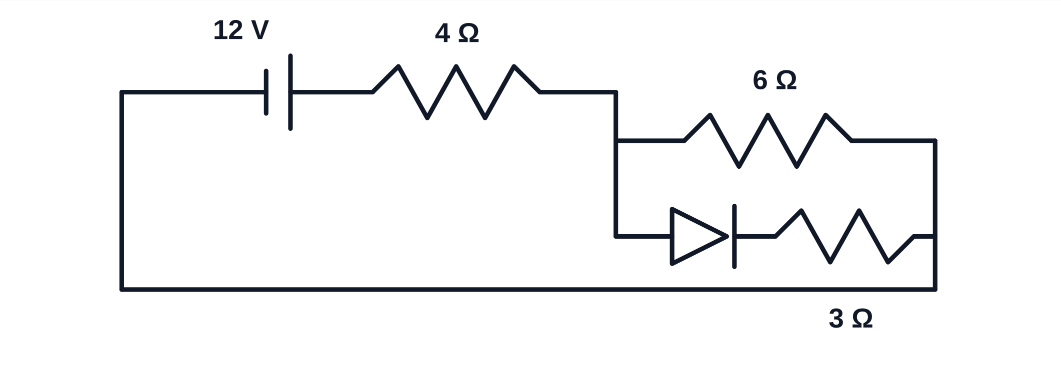
Diodes and equivalent resistance

Answer **B**

Question 6

An ideal $12\ \mathrm{V}$ source supplies a $4\,\Omega$ resistor in series with two parallel branches. One branch is $6\,\Omega$; the other contains an ideal diode in series with $3\,\Omega$. With the first polarity the diode conducts. When the source polarity is reversed, the diode blocks. What is the ratio

$$\frac{\text{power drawn with the diode conducting}}{\text{power drawn with the diode blocking}}?$$



A $\frac{2}{3}$

B $\frac{5}{3}$

C $\frac{3}{2}$

D $\frac{5}{2}$

E $\frac{10}{3}$

WORKED SOLUTION

When the diode conducts, $6\,\Omega$ and $3\,\Omega$ are in parallel, giving $2\,\Omega$. Total resistance is $6\,\Omega$, so $P_{\text{on}} = V^2/R = 144/6 = 24\ \mathrm{W}$.

When it blocks, total resistance is $4 + 6 = 10\,\Omega$, so $P_{\text{off}} = 144/10 = 14.4\ \mathrm{W}$. Therefore $P_{\text{on}}/P_{\text{off}} = 24/14.4 = 5/3$. The correct option is **B.**

P-07

Radioactive data and uncertainty

Answer **C**

Question 7

A Geiger counter records the following counts in repeated $30\ \mathrm{s}$ intervals. The source and detector remain fixed.

Measurement	1	2	3	4
Source + background	148	161	153	139
Background only	18	22	16	24

Which statements are correct?

1. Repeated counts can differ even if the source activity and apparatus are unchanged.
2. The best estimate from these data of the source-only count in $30\ \mathrm{s}$ is 130.25.
3. Subtracting the mean background removes all random uncertainty from the source estimate.

A 1 only

B 2 only

C 1 and 2 only

D 2 and 3 only

E 1, 2 and 3

WORKED SOLUTION

Radioactive decay is random, so statement 1 is correct. The mean source-plus-background count is $(148 + 161 + 153 + 139)/4 = 150.25$, while the mean background is $(18 + 22 + 16 + 24)/4 = 20$. Their difference is 130.25, so statement 2 is correct.

Background subtraction corrects the central estimate but does not eliminate fluctuations in either set of counts, so statement 3 is false. The correct option is **C.**

P-08

Newton's laws and coupled motion

Answer A

Question 8

A $1200\ \mathrm{kg}$ car pulls an $800\ \mathrm{kg}$ trailer along a horizontal road. The resistive forces on the car and trailer are $400\ \mathrm{N}$ and $200\ \mathrm{N}$, respectively. Both accelerate forwards at $1.5\ \mathrm{m\ s^{-2}}$. What are the driving force on the car and the force exerted by the trailer on the car?

- **A** $3600\ \mathrm{N}$ forwards; $1400\ \mathrm{N}$ backwards
- **B** $3000\ \mathrm{N}$ forwards; $1200\ \mathrm{N}$ backwards
- **C** $3600\ \mathrm{N}$ forwards; $1200\ \mathrm{N}$ backwards
- **D** $3400\ \mathrm{N}$ forwards; $1400\ \mathrm{N}$ backwards
- **E** $3600\ \mathrm{N}$ forwards; $1400\ \mathrm{N}$ forwards

WORKED SOLUTION

For the complete car-trailer system, if the driving force is F, then

$$F - (400 + 200) = (1200 + 800)(1.5),$$

so $F = 3600\ \mathrm{N}$. For the trailer alone, if the coupling force is T,

$$T - 200 = 800(1.5),$$

giving $T = 1400\ \mathrm{N}$. By Newton's third law, the trailer exerts an equal $1400\ \mathrm{N}$ force backwards on the car. The correct option is **A.**

P-09

Doppler effect from wavefront spacing

Answer B

Question 9

A sound source emits $240\ \mathrm{Hz}$ while moving directly towards a stationary observer at $60\ \mathrm{m\ s^{-1}}$. The speed of sound in the still air is $360\ \mathrm{m\ s^{-1}}$. During one source period, a crest travels at the speed of sound while the source moves closer to the preceding crest. What frequency does the observer detect?

- **A** $270\ \mathrm{Hz}$
- **B** $288\ \mathrm{Hz}$
- **C** $240\ \mathrm{Hz}$
- **D** $300\ \mathrm{Hz}$
- **E** $320\ \mathrm{Hz}$

WORKED SOLUTION

One source period is $T = 1/240\ \mathrm{s}$. In this time the earlier crest travels $360T$, while the source advances $60T$. The wavelength in front of the source is therefore

$$\lambda = (360 - 60)T = \frac{300}{240} = 1.25\ \mathrm{m}.$$

The stationary observer receives waves travelling at $360\ \mathrm{m\ s^{-1}}$, so

$$f_{\mathrm{obs}} = \frac{360}{1.25} = 288\ \mathrm{Hz}.$$

The correct option is **B.**

P-10

Transformer ratio and efficiency

Answer **D**

Question 10

A transformer has $240\ \mathrm{V}$ across its primary and $12\ \mathrm{V}$ across its secondary. A $6\ \Omega$ resistor is connected across the secondary. The transformer is 80% efficient, where efficiency is secondary output power divided by primary input power. Assume that its terminal voltage ratio still obeys $V_s/V_p = N_s/N_p$. What are the primary current I_p and the turns ratio N_s/N_p?

A $I_p = \frac{1}{10}\ \mathrm{A},\ N_s/N_p = \frac{1}{20}$

B $I_p = \frac{1}{8}\ \mathrm{A},\ N_s/N_p = \frac{1}{10}$

C $I_p = \frac{1}{10}\ \mathrm{A},\ N_s/N_p = \frac{1}{10}$

D $I_p = \frac{1}{8}\ \mathrm{A},\ N_s/N_p = \frac{1}{20}$

E $I_p = \frac{1}{5}\ \mathrm{A},\ N_s/N_p = \frac{1}{20}$

WORKED SOLUTION

The secondary current is $12/6 = 2\ \mathrm{A}$, so output power is $12 \times 2 = 24\ \mathrm{W}$. Input power is $24/0.80 = 30\ \mathrm{W}$, giving $I_p = 30/240 = 1/8\ \mathrm{A}$. Also

$$\frac{N_s}{N_p} = \frac{V_s}{V_p} = \frac{12}{240} = \frac{1}{20}.$$

The correct option is **D.**

P-11

Two-stage ideal-gas change

Answer **C**

Question 11

A fixed mass of ideal gas initially has pressure p, volume V, absolute temperature T and density ρ. It is first compressed isothermally to volume $V/2$. Keeping this new volume fixed, it is then heated until its absolute temperature is $3T/2$. What are its final pressure and density? For this question, use $pV/T = \text{constant}$ for a fixed mass of ideal gas.

A $\frac{3}{2}p,\ 2\rho$

B $2p,\ \frac{3}{2}\rho$

C $3p,\ 2\rho$

D $3p,\ 3\rho$

E $4p,\ 2\rho$

WORKED SOLUTION

During the isothermal compression, pV is constant. Halving the volume therefore doubles the pressure to $2p$. At fixed volume, pressure is proportional to absolute temperature, so the second step multiplies the pressure by $3/2$:

$$p_{\text{final}} = 2p\left(\frac{3}{2}\right) = 3p.$$

The mass is unchanged and the final volume is $V/2$, so $\rho_{\text{final}} = m/(V/2) = 2\rho$. The correct option is **C.**

P-12

Terminal velocity and drag

Answer **E**

Question 12

An object falls vertically through a fluid. Its weight is W, its upthrust is a constant U, and its drag force is kv^2, where v is its speed. Its terminal speed is v_t. What is the resultant force when its speed is $v_t/2$?

- **A** $\frac{1}{4}(W-U)$ downward
- **B** $\frac{1}{2}(W-U)$ downward
- **C** $\frac{3}{4}(W-U)$ upward
- **D** $\frac{1}{4}(W-U)$ upward
- **E** $\frac{3}{4}(W-U)$ downward

WORKED SOLUTION

At terminal speed, $kv_t^2 = W - U$. At $v_t/2$, the drag is $\frac{1}{4}kv_t^2 = \frac{1}{4}(W-U)$. The downward resultant is therefore

$$(W-U) - \frac{1}{4}(W-U) = \frac{3}{4}(W-U).$$

The correct option is **E.**

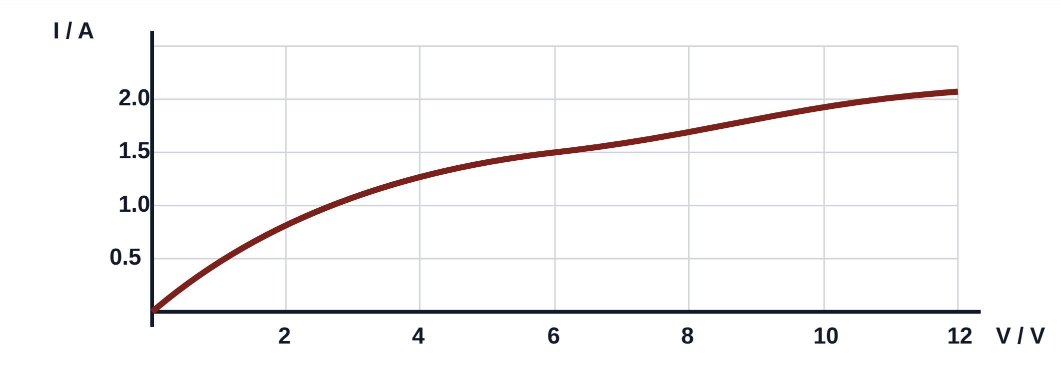
P-13

Non-ohmic load-line reasoning

Answer **C**

Question 13

The graph shows the current-voltage characteristic of a filament lamp. The lamp is connected in series with a $4\,\Omega$ resistor across an ideal $12\ \mathrm{V}$ source. What power is dissipated in the lamp?



A $4.5\ \mathrm{W}$

B $6\ \mathrm{W}$

C $9\ \mathrm{W}$

D $12\ \mathrm{W}$

E $18\ \mathrm{W}$

WORKED SOLUTION

If the lamp voltage is V_L and the current is I, the series resistor has voltage $4I$, so

$$V_L + 4I = 12.$$

From the graph, the lamp has $I = 1.5\ \mathrm{A}$ at $V_L = 6\ \mathrm{V}$; this satisfies $6 + 4(1.5) = 12$. The increasing lamp curve and decreasing load line give a unique operating point. Thus $P_L = V_L I = 6(1.5) = 9\ \mathrm{W}$. The correct option is **C.**

P-14

Comparative heating data

Answer **C**

Question 14

Equal masses of solids A and B, both initially at $20^\circ\mathrm{C}$, are heated by identical constant-power heaters. Energy loss is negligible. The results are:

Solid	Melting temperature	Time to reach melting temperature	Time at constant temperature while melting
A	$60^\circ\mathrm{C}$	$40\ \mathrm{s}$	$50\ \mathrm{s}$
B	$80^\circ\mathrm{C}$	$60\ \mathrm{s}$	$30\ \mathrm{s}$

Which statements are correct?

1. Solid B has the higher melting point.
2. In the solid state, A has the greater specific heat capacity.
3. A has the greater specific latent heat of fusion.

- **A** 1 only
- **B** 2 only
- **C** 1 and 3 only
- **D** 2 and 3 only
- **E** 1, 2 and 3

WORKED SOLUTION

The melting temperatures make statement 1 true. Before melting, $Pt = mc\Delta T$. For A, $t/\Delta T = 40/40 = 1\ \mathrm{s\,K^{-1}}$; for B, $60/60 = 1\ \mathrm{s\,K^{-1}}$. The masses and powers are equal, so the specific heat capacities are equal and statement 2 is false.

During melting, $Pt = mL$. A remains on the plateau for longer, so A has the greater latent heat and statement 3 is true. The correct option is **C.**

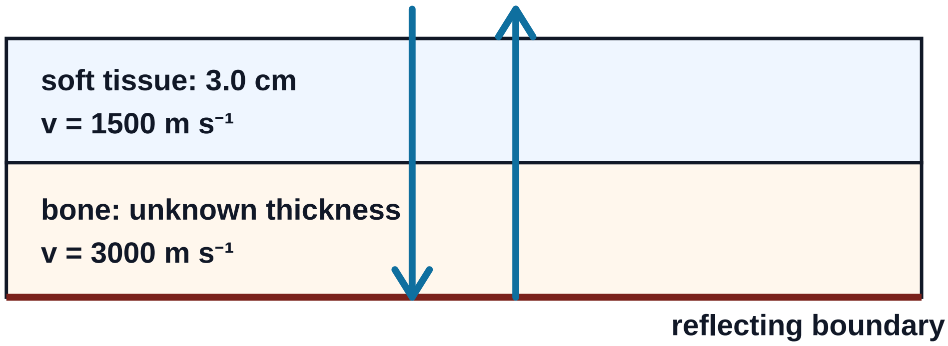
P-15

Ultrasound echo through two media

Answer **A**

Question 15

An ultrasound pulse travels normally through $3.0\ \mathrm{cm}$ of soft tissue at $1500\ \mathrm{m\ s^{-1}}$, then through bone at $3000\ \mathrm{m\ s^{-1}}$. It reflects from a boundary within the bone and returns along the same path. The echo is received $60\ \mu\mathrm{s}$ after emission. How far below the surface is the reflecting boundary?



A $6.0\ \mathrm{cm}$

B $3.0\ \mathrm{cm}$

C $4.5\ \mathrm{cm}$

D $7.5\ \mathrm{cm}$

E $9.0\ \mathrm{cm}$

WORKED SOLUTION

The round-trip time in the soft tissue is

$$t_s = \frac{2(0.030)}{1500} = 40\ \mu\mathrm{s}.$$

The remaining $20\ \mu\mathrm{s}$ is the round-trip time in bone. If the one-way bone thickness is x,

$$\frac{2x}{3000} = 20 \times 10^{-6} \quad \Rightarrow \quad x = 0.030\ \mathrm{m} = 3.0\ \mathrm{cm}.$$

The reflecting boundary is therefore $3.0 + 3.0 = 6.0\ \mathrm{cm}$ below the surface. The correct option is **A.**

P-16

Energy rate and fluid motion

Answer **C**

Question 16

An electric pump takes in power at $500\ \mathrm{W}$ and is 80% efficient. It pumps water at $2.0\ \mathrm{kg\ s^{-1}}$ from a large reservoir to an outlet $15\ \mathrm{m}$ higher. The reservoir surface and outlet are both at atmospheric pressure, and the speed of the reservoir surface is negligible. Neglect all losses other than those represented by the efficiency, and take $g = 10\ \mathrm{N\ kg^{-1}}$.

What is the speed of the water as it leaves the outlet?

- **A** $5\ \mathrm{m\ s^{-1}}$
- **B** $5\sqrt{2}\ \mathrm{m\ s^{-1}}$
- **C** $10\ \mathrm{m\ s^{-1}}$
- **D** $10\sqrt{2}\ \mathrm{m\ s^{-1}}$
- **E** $20\ \mathrm{m\ s^{-1}}$

WORKED SOLUTION

The useful power is $0.80(500) = 400\ \mathrm{W}$. Raising $2.0\ \mathrm{kg}$ each second through $15\ \mathrm{m}$ requires gravitational power

$$\dot{m}gh = 2(10)(15) = 300\ \mathrm{W}.$$

The remaining $100\ \mathrm{W}$ becomes kinetic energy of the water each second:

$$\frac{1}{2}\dot{m}v^2 = 100 \quad \Rightarrow \quad \frac{1}{2}(2)v^2 = 100.$$

Thus $v = 10\ \mathrm{m\ s^{-1}}$. The correct option is **C.**

P-17

Electromagnetic induction

Answer **E**

Question 17

The north pole of a bar magnet moves directly towards one end of a coil at speed u, is stopped, and is then pulled away. As it passes the same positions on the way out, its speed is $u/2$. Which statements are correct?

1. While the north pole approaches, the near face of the coil behaves as a north pole.
2. While the magnet is stationary, the induced potential difference is zero.
3. While the magnet recedes through the same positions, the induced potential difference has the opposite direction and a smaller magnitude than during approach.

- **A** 1 only
- **B** 2 only
- **C** 1 and 2 only
- **D** 2 and 3 only
- **E** 1, 2 and 3

WORKED SOLUTION

Lenz's law makes the near face repel the approaching north pole, so it acts north. No flux change occurs while the magnet is stationary. On withdrawal, the flux change reverses, reversing the induced voltage; at the same position its rate of change is smaller because the speed is halved. All statements are correct, so the answer is **E.**

P-18

Radiation absorption and count rates

Answer **B**

Question 18

A source emits alpha, beta-minus and gamma radiation. A detector has a background rate of $20 \text{ counts min}^{-1}$. It records:

- $620 \text{ counts min}^{-1}$ with no absorber;
- $420 \text{ counts min}^{-1}$ after paper is inserted;
- $170 \text{ counts min}^{-1}$ after aluminium is added behind the paper;
- $20 \text{ counts min}^{-1}$ after thick lead is added.

Assume the paper removes only alpha, the aluminium then removes only beta-minus, and the lead removes the remaining gamma. Which option gives, respectively, the ratio of the source's beta-minus count rate to its gamma count rate and the fraction of the source-only count rate due to alpha radiation?

A $2:1;\ \frac{1}{3}$

B $5:3;\ \frac{1}{3}$

C $3:2;\ \frac{10}{31}$

D $5:4;\ \frac{1}{2}$

E $3:5;\ \frac{1}{3}$

WORKED SOLUTION

The beta-minus component is the decrease when aluminium is added:

$$420 - 170 = 250 \text{ counts min}^{-1}.$$

After paper and aluminium, the source contribution is the gamma rate. Subtracting background gives

$$170 - 20 = 150 \text{ counts min}^{-1}.$$

The alpha component is the decrease when paper is inserted, $620 - 420 = 200 \text{ counts min}^{-1}$. The total source-only rate is $620 - 20 = 600 \text{ counts min}^{-1}$, so the alpha fraction is $200/600 = 1/3$.

Thus the required pair is $5:3;\ 1/3$. The correct option is **B.**

P-19

Hydrostatic pressure and force

Answer **B**

Question 19

A vertical-sided container of base area A holds an oil layer of density ρ and depth $2h$ above a liquid of density 2ρ and depth h. The same volumes of both liquids are transferred to a second vertical-sided container of base area $2A$. Compared with the first container, how do the gauge pressure and the total force due to liquid pressure on the horizontal base change?

- **A** Pressure halves; force halves
- **B** Pressure halves; force is unchanged
- **C** Pressure is unchanged; force doubles
- **D** Pressure becomes one quarter; force halves
- **E** Pressure doubles; force is unchanged

WORKED SOLUTION

Initially, $p_1 = \rho g(2h) + 2\rho g(h) = 4\rho gh$. Doubling base area halves each depth, so

$$p_2 = \rho g(h) + 2\rho g\left(\frac{h}{2}\right) = 2\rho gh = \frac{p_1}{2}.$$

But $F_1 = p_1 A = 4\rho ghA$, while $F_2 = p_2(2A) = 4\rho ghA$. Pressure halves and force is unchanged. The correct option is **B.**

P-20

Electrostatic induction and earthing

Answer **C**

Question 20

A positively charged rod is brought close to an initially neutral isolated metal sphere without touching it. While the rod remains nearby, the sphere is briefly connected to Earth. The Earth connection is removed before the rod is taken away. What is the final charge on the sphere, and what happened while it was earthed?

- **A** Neutral; equal numbers of electrons flowed in both directions
- **B** Positive; electrons flowed from Earth to the sphere
- **C** Negative; electrons flowed from Earth to the sphere
- **D** Positive; electrons flowed from the sphere to Earth
- **E** Negative; positive charge flowed from Earth to the sphere

WORKED SOLUTION

The positive rod attracts electrons towards the sphere. While earthed, additional electrons flow from Earth onto it. Removing the Earth connection traps this excess; after the rod is removed, the electrons redistribute and the sphere remains negative. The correct option is **C.**

P-21

Thermal radiation and energy rates

Answer **D**

Question 21

Two otherwise identical hot metal containers are at the same temperature in identical cooler surroundings. The initial net radiative loss from the dull black container is four times that from the polished silver container. Each also loses energy by non-radiative processes at the same rate H. The measured initial total loss rates are $70\ \mathrm{W}$ for the black container and $40\ \mathrm{W}$ for the silver container. What fraction of the black container's total loss is by radiation?

- **A** $\frac{3}{7}$
- **B** $\frac{1}{2}$
- **C** $\frac{3}{5}$
- **D** $\frac{4}{7}$
- **E** $\frac{5}{7}$

WORKED SOLUTION

Let the black container's radiative loss be R. The silver container's radiative loss is then $R/4$. Hence

$$R + H = 70, \qquad \frac{R}{4} + H = 40.$$

Subtracting gives $3R/4 = 30$, so $R = 40\ \mathrm{W}$. The required fraction is therefore

$$\frac{R}{70} = \frac{40}{70} = \frac{4}{7}.$$

The correct option is **D.**

P-22

Momentum and kinetic energy

Answer **E**

Question 22

A particle of mass m travels at speed $3u$. It collides with a particle of mass $2m$ travelling in the same direction at speed u. They stick together. Which option gives their common speed and the fraction of the initial kinetic energy that is lost?

- **A** $\frac{5u}{3}$; $\frac{8}{25}$
- **B** $\frac{4u}{3}$; $\frac{1}{4}$
- **C** $\frac{5u}{3}$; $\frac{4}{33}$
- **D** $\frac{4u}{3}$; $\frac{1}{3}$
- **E** $\frac{5u}{3}$; $\frac{8}{33}$

WORKED SOLUTION

Momentum conservation gives $3mu + 2mu = 3mv$, so $v = 5u/3$. Initial kinetic energy is $\frac{1}{2}m(3u)^2 + \frac{1}{2}(2m)u^2 = \frac{11}{2}mu^2$. Final kinetic energy is

$$\frac{1}{2}(3m)\left(\frac{5u}{3}\right)^2 = \frac{25}{6}mu^2.$$

The loss is $\frac{4}{3}mu^2$, so the fraction lost is $(4/3)/(11/2) = 8/33$. The required pair is therefore $5u/3;\ 8/33$, and the correct option is **E.**

P-23

Waves in a material

Answer **C**

Question 23

In medium X, a wave has speed v, wavelength λ and period T. It enters normally into a slab of medium Y, where its speed is $3v/4$. The slab thickness is 6λ. How many wavelengths fit across the slab, and how long does the wave take to cross it?

- **A** 6 wavelengths; $6T$
- **B** 8 wavelengths; $6T$
- **C** 8 wavelengths; $8T$
- **D** 6 wavelengths; $8T$
- **E** 9 wavelengths; $8T$

WORKED SOLUTION

Frequency is unchanged, so the wavelength in Y is $\lambda_Y = (3/4)\lambda$. Therefore

$$\frac{6\lambda}{(3/4)\lambda} = 8$$

wavelengths fit in the slab. Crossing time is $6\lambda/(3v/4) = 8\lambda/v = 8T$, because $T = \lambda/v$. The correct option is **C.**

P-24

Multi-stage power transmission

Answer **B**

Question 24

In an original transmission system, power P enters cables of total resistance R at voltage V. The power lost in the cables is exactly $P/5$.

In a redesigned system, the same generator power P enters a transformer of efficiency 80%, whose output sends power through the same cables at voltage $4V$. At the far end, a second transformer of efficiency 75% supplies a load.

What fraction of P reaches the load?

- **A** $\frac{3}{5}$
- **B** $\frac{297}{500}$
- **C** $\frac{99}{125}$
- **D** $\frac{12}{25}$
- **E** $\frac{99}{100}$

WORKED SOLUTION

Originally $I = P/V$ and $I^2R = P/5$. The first transformer sends $4P/5$ into the cables at $4V$, so the new cable current is

$$I' = \frac{4P/5}{4V} = \frac{P}{5V} = \frac{I}{5}.$$

The new cable loss is therefore $(I/5)^2R = (P/5)/25 = P/125$. The power entering the receiving transformer is

$$\frac{4P}{5} - \frac{P}{125} = \frac{99P}{125}.$$

Its 75% efficiency gives load power $(3/4)(99P/125) = 297P/500$. The correct option is **B.**

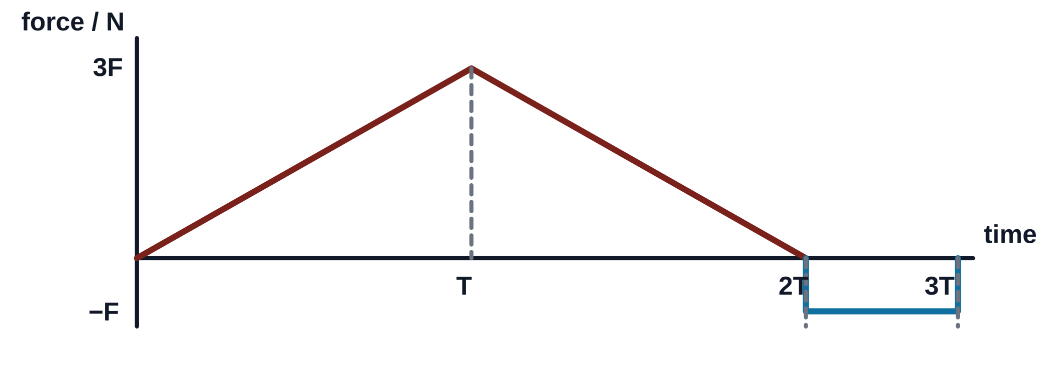
P-25

Signed impulse, collision and energy

Answer **D**

Question 25

A trolley of mass m is initially at rest. The horizontal force on it varies with time as shown. Immediately after $t = 3T$, it collides with a stationary trolley of mass $3m$; the trolleys stick together. They then move up a smooth ramp and come momentarily to rest at vertical height h.



Which expression gives h?

A $\dfrac{F^2T^2}{2m^2g}$

B $\dfrac{F^2T^2}{4m^2g}$

C $\dfrac{F^2T^2}{32m^2g}$

D $\dfrac{F^2T^2}{8m^2g}$

E $\dfrac{9F^2T^2}{32m^2g}$

WORKED SOLUTION

The positive triangular area is $\frac{1}{2}(2T)(3F) = 3FT$; the negative rectangular area is $-FT$. The net impulse is therefore $2FT$, so the first trolley's momentum just before collision is $2FT$.

Momentum conservation in the sticking collision gives the combined mass $4m$ speed

$$v = \frac{2FT}{4m} = \frac{FT}{2m}.$$

On the smooth ramp, $\frac{1}{2}(4m)v^2 = (4m)gh$, so $h = v^2/(2g) = F^2T^2/(8m^2g)$. The correct option is **D.**

P-26

Force and instantaneous power

Answer **A**

Question 26

A lift of mass m is moving vertically upward. At three instants X, Y and Z, its upward speed has the same value v. Its acceleration is upward with magnitude a at X, zero at Y, and downward with magnitude a at Z, where $0 < a < g$. The cable remains taut and resistive forces are negligible.

Which ordering is correct for the instantaneous powers P_X, P_Y, P_Z delivered to the lift by the cable?

A $P_X > P_Y > P_Z$

B $P_Z > P_Y > P_X$

C $P_X = P_Y = P_Z$

D $P_X > P_Z > P_Y$

E $P_Y > P_X > P_Z$

WORKED SOLUTION

Newton's second law gives cable tensions $m(g+a)$, mg and $m(g-a)$ at X, Y and Z, respectively. At each instant the cable force and velocity are parallel and the speed is the same, so $P = Tv$.

Therefore $P_X > P_Y > P_Z$, and the correct option is **A.**

P-27

Nuclear transformations and neutron number

Answer **E**

Question 27

A nucleus initially has mass number A, proton number Z and neutron number $N = A - Z$. It emits some alpha particles and some beta-minus particles. Its final mass number is $A - 12$, and its final neutron number is $N - 8$. How many emissions of each type occurred, and what is the final proton number?

A 3 alpha, 2 beta-minus; $Z - 8$

B 3 alpha, 4 beta-minus; $Z - 2$

C 2 alpha, 2 beta-minus; $Z - 2$

D 3 alpha, 2 beta-minus; $Z - 6$

E 3 alpha, 2 beta-minus; $Z - 4$

WORKED SOLUTION

Only alpha decay changes mass number, by -4 each time. A change from A to $A - 12$ therefore requires three alpha decays. These reduce neutron number by 2 each, taking N to $N - 6$.

Each beta-minus decay converts a neutron into a proton, so it reduces neutron number by 1. Reaching $N - 8$ therefore requires two beta-minus decays. The proton number changes by

$$-2(3) + 1(2) = -4,$$

so the final proton number is $Z - 4$. The correct option is **E.**